

Human Subjects Review and Institutional Review Boards

05-499/899 Fall 2024

Celebrating Accessibility

<https://cmu-05-499.github.io>

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Administrivia

- Jeffrey Ruffing from Inglis will visit on Monday, November 2 to talk about DIY Assistive Technology using 3D printing and electronics.
- Are you having any difficulty connecting with your partner organization? See me after class.

Research with Human Subjects

Human Subjects Research

- Many different goals
 - Find out more about human behavior
 - Check the quality or efficacy of software / device / product
 - Compare software / device / product
- Important factors
 - Research should be ethical
 - How to get the right information
 - How to measure the right information
 - How to analyze the right information

Ethical Consideration

- Sometimes tests can be **distressing**
- You have a **responsibility to alleviate that stress**
 - make voluntary with informed consent
 - avoid pressure to participate
 - let users know they can stop at any time
 - offer breaks
 - stress you are testing the system, not the user
 - make collected data as anonymous as possible
- Often must get **human subjects approval**

What is an Institutional Review Board? Why do we need it?

The Readings

1. The 1979 Belmont Report
2. Institutional review board (IRB) and ethical issues in clinical research
3. The Milgram Experiment: Obedience to Authority

(Too Many) Examples of Ethical Research Failures

Tuskegee Study 1930s-1972

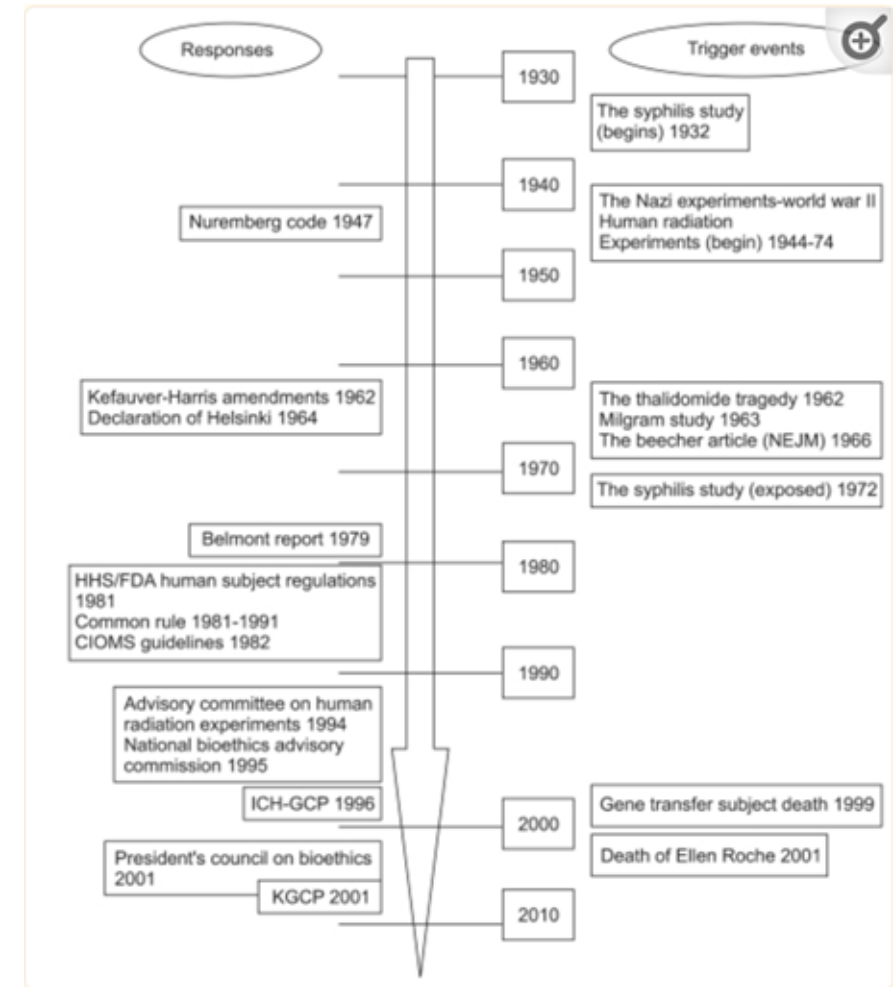
Nazi Experimentation

Nuremberg Trials 1947

Thalidomide 1957-1961

Milgram Study 1960s

Stanford Prison Experiment 1971



Tuskegee Syphilis Study (1932-1972)

US Public Health Service (PHS) ran study on syphilis for 40 years!

399 low-income black men told they were being treated for “bad blood.”

They actually had syphilis and were left untreated

Local physicians given lists of people not to treat

Original study only meant to last 6-9 months



<https://www.britannica.com/event/Tuskegee-syphilis-study>

Tuskegee Syphilis Study (1932-1972)

Initially no syphilis treatment available, but by 1947, penicillin, the standard treatment, was withheld

Men died. Their families got infected.

For participating in the study, the men were given free medical exams, free meals and transportation, and free burial insurance

Stopped in 1972 after PHS employees leaked info to the press

"I don't know what they used us for. I ain't never understood the study." - a survivor

Thalidomide

- Originally developed as a tranquilizer in 1957 in Europe.
 - Marketed for anxiety and *morning sickness*.
- By 1961, was found to have caused severe birth defects
 - Shortened limbs (dysmelia), malformed arms and legs (phocomelia), missing limbs (amelia)
 - Malformed ears and hearing loss, eye malformations, etc.
 - 30-60% of newborns died (likely heart malformations)
- Severity and location of birth defects vary based on timing of exposure
 - 20-36 days after fertilization (34-50 days after last menstrual cycle)
- **Was never tested on women who became pregnant!**

Milgram Obedience Experiment (1960s)

Before the establishment of IRBs

Very distressing to participants, significant deception involved



Zimbardo Prison Experiment (1971)

- Stanford professor Philip Zimbardo
- Office of Naval Research-funded study of the psychological effects of being a prisoner or prison guard.
 - Studied the power of roles, rules, group identity and the situational validation of behavior that would repulse ordinary people.
- Recruited 24 SF Bay Area male college students to participate in a 2-week long prison life simulation.
- Randomly chosen to be prisoners or guards
 - Prisoners confined to jail cells and the prison yard.
 - Guards were given uniforms, batons, mirrored sunglasses, and told to maintain law and order.
- Experiment lasted six days before being ended early after extreme psychological abuse and brutality of “prisoners” by “guards.”

Zimbardo Prison Experiment (1971)



Prison Experiment Impact

- Participants
 - Experienced long-term psychological trauma
- Science
 - People conform to social pressures. They alter their behavior to match group stereotypes.
 - Adolf Eichmann claims he bore no responsibility for organizing the Holocaust because he was simply "doing his job." He did his duty, obeyed orders, and obeyed the law.
- Society
 - Juveniles no longer housed with adult prisoners before trial due to risk of violence against them.
 - Zimbardo was an expert witness for the court-martial an Abu Ghraib prison guard for torture and abuse.
 - Despite much evidence, we have difficulty believing in "the banality of evil."

National Research Act (1974)

- 1974 National Research Act (Pub. L. 93-348)
 - Direct response to whistleblowing ending the Tuskegee study
- Established the “National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research”
 - Identify basic ethical principles
Develop guidelines to assure research is carried out according to those principles
- **Required IRBs** at institutions receiving support for human subjects research for Health, Education, and Welfare within Dept. of Health and Human Services
- February 1976 - Held four-day conference at Smithsonian Institution’s Belmont Conference Center

Belmont Report (1979)

3 core principles

1. **Respect for persons** - protecting the autonomy of human subjects
2. **Beneficence** - protecting participants from harm
3. **Justice** - protecting human subjects from exploitation

Participation Activity

- Pair up with a neighbor.
- Write a post to **#in-class** about the Zimbardo Prison Experiment.
- Answer these three questions:
 1. How did it violate respect for persons?
 2. How did it fail to protect participants from harm?
 3. How did it fail to protect human subjects from exploitation?

Federal Regulations

Basic Ethical Principles

Respect for Persons

Beneficence

Justice



Applications

Informed Consent

Assessment of Risk and Benefits

Minimize Risk

Protect vulnerable

Fair selection of Subjects

System of Institutional Review Boards (IRBs) to monitor human subject research

Federal regulations for treatment of human subjects

<http://ohrp.osophs.dhhs.gov/humansubjects/guidance/45cfr46.htm>

General Process - Designing a Study

1. Define the **goals** for your study
 - What do you want to learn?
2. Choose participants
3. Define the **procedures/tasks** for each group of participants
 - What will you ask participants to do?
4. What kind of **data** do you need to collect?

General Process - Designing a Study

- Participant Recruitment
- Data Collection
 - Questionnaires - Pre/Post Test
 - Recordings
- Compensation
- Risks/Benefits
 - Protecting the Participants
 - Protecting the Data

General Process – Informed Consent

- Providing enough information
- Ensuring the information is understood
- Documenting voluntary consent to participate

Facebook “Mood Manipulation Study” (2012)

- “Facebook’s data scientists manipulated the News Feeds of **689,003 users**, **removing either all of the positive posts or all of the negative posts** to see how it affected their moods. If there was a week in January 2012 where you were only seeing photos of dead dogs or incredibly cute babies, you may have been part of the study.”
- “The researchers ... found that **emotions were contagious**. When positive expressions were reduced, people produced fewer positive posts and more negative posts; when negative expressions were reduced, the opposite pattern occurred”
- Automated testing “**was consistent with Facebook’s Data Use Policy**, to which all users agree prior to creating an account on Facebook, constituting informed consent for this research.”

<http://www.forbes.com/sites/kashmirhill/2014/06/28/facebook-manipulated-689003-users-emotions-for-science/>

<http://www.forbes.com/sites/gregorymcneal/2014/06/30/controversy-over-facebook-emotional-manipulation-study-grows-as-timeline-becomes-more-clear>

Participation Activity

- Pair up with a neighbor.
- Was Facebook's "Mood Manipulation Study" Ethical?
 - Post 2 reasons to **#in-class** for why it should be considered ethical.
 - Post 2 reasons to **#in-class** why it should NOT be considered ethical.
 - Tag your partner!
- Pair up with another pair and discuss your reasons.
 - Can you agree on whether it was an ethical study?